
Smart AI Business Assistant Platform

Assessment Task Brief for Intern Evaluation

Project Category	AI Agent, Automation, RAG, and Analytics Platform
Target Users	Small and medium businesses (SMEs)
Submission Type	Production-oriented MVP with backend, dashboard, and deployment readiness
Assessment Focus	Architecture, implementation quality, reliability, and business readiness

Build a real client-ready MVP for an AI-powered business assistant that can handle customer questions, capture and manage leads, automate routine work, remember context over time, retrieve answers from business documents, and provide admin-level visibility through analytics and operational logs. The final submission must look and behave like a deployable product, not a demo chatbot.

Evaluation Goal

This assignment is designed to measure practical system design, multi-agent orchestration, reliability engineering, API structure, and dashboard implementation. Teams should show disciplined engineering choices, clean code organization, and strong product thinking.

Expected Output

A polished working prototype with documentation, deployment support, and a clear demonstration of how an SME can use the system in day-to-day operations.

1. Project Overview

The proposed platform should function as a unified business operations assistant for SMEs. It must be capable of responding to business queries, creating and qualifying leads, running automated workflows, managing memory across conversations, retrieving knowledge from uploaded documents, and exposing all major activity inside an admin dashboard.

2. Mandatory Functional Requirements

Module	Required capability
AI Assistant	Answer business-related questions, support multi-turn conversation, stay contextual, and avoid unsupported or hallucinated answers.
RAG System	Provide document upload, semantic chunking, embeddings, and vector retrieval using a supported vector store.
Memory Handling	Include short-term conversation memory, long-term user memory, and personalized interaction behavior.
Lead Capture	Capture lead details naturally, classify hot/warm/cold leads, store history, and support follow-up generation.
Workflow Automation	Implement at least three useful automations such as email summarization, CRM/Sheets sync, response flow, or calendar booking.
Multi-Agent Workflow	Use at least Planner, Executor, and Validator/Critic agents with clear orchestration.
Backend APIs	Build the backend with FastAPI, modular structure, and async endpoints where appropriate.
Dashboard	Provide lead analytics, conversation logs, workflow logs, AI usage metrics, and document management.
Authentication	Include login/signup, session handling, and role-based access where possible.
Error Handling	Add retry logic, fallback responses, structured logging, and failure notifications.
Deployment	Provide Docker support, environment variable handling, and a clear README setup guide.

3. Workflow Scope

The assistant should support the complete business flow from first interaction to follow-up and reporting. A user should be able to upload knowledge documents, interact with the assistant, capture a lead, trigger automations, and review outcomes from the dashboard.

4. Bonus Features

- Voice interaction
- Multimodal input
- Streaming responses
- Redis caching
- Autonomous retries
- Webhooks
- Drag-and-drop workflow builder
- AI evaluation system
- Agent tracing or monitoring

5. Suggested Technical Stack

Layer	Suggested choices
Backend	FastAPI, Python, async services
Agent orchestration	LangGraph, CrewAI, or custom orchestration
Vector store	ChromaDB, FAISS, or Pinecone
Database	PostgreSQL or MongoDB for persistent records
Auth	JWT or session-based authentication
Deployment	Docker and environment-based configuration

6. Deliverables Expected from Each Intern

- Source code with modular project structure
- Working backend APIs
- Functional assistant with RAG and memory
- Lead capture and automation features
- Dashboard with operational visibility
- Dockerized setup and environment configuration
- README with installation and run instructions

7. Assessment Rubric

Criterion	Weight	What is evaluated
Core functionality	30%	Completion of mandatory features and working end-to-end flow.
Architecture and code quality	20%	Clean structure, modularity, maintainability, and API design.
AI and RAG quality	20%	Response quality, retrieval design, memory handling, and hallucination control.
Automation and agent workflow	15%	Useful automations, orchestration, and reliability of the workflow.
Dashboard and analytics	10%	Clarity of reporting, logs, and administrative visibility.
Deployment and documentation	5%	Docker, env handling, setup guide, and presentation readiness.

8. Submission Notes

- The final solution should be stable enough to demonstrate to a client or stakeholder.
- All important modules should be connected and not isolated as separate proof-of-concepts.
- The interface should prioritize practicality, business usefulness, and clear operational visibility.
- Any assumptions, third-party dependencies, or limitations should be documented clearly in the README.

Assessment Focus: build something that can credibly support a small business workflow from conversation to automation to reporting.